

# TypeScript & Playwright Workshop

## Three-Day Course Outline

Classroom GitHub Repo: [https://github.com/yithril/Dexian\\_TypescriptAndPlayWright](https://github.com/yithril/Dexian_TypescriptAndPlayWright)

### Day 1: TypeScript Fundamentals

**Target:** Be comfortable reading and writing the TypeScript you'll see in Playwright tests.

#### Topics Covered

- What TypeScript is and why it is useful
- Setting up a TypeScript project
- Basic type annotations
- Writing and typing functions
- Union types
- Interfaces for objects
- Structural typing
- Enums

#### Lab

Hands-on practice writing TypeScript interfaces, typed arrays, and functions with typed parameters and return values.

### Day 2: Introduction to Playwright

**Target:** Understand how a Playwright test is built, walk through real examples together, and introduce the shared Playwright lab.

#### Topics Covered

- Review of the Day 1 lab
- What Playwright is and why it is useful for test automation
- Setting up a Playwright project
- Anatomy of a Playwright test
- Locators and selectors
- Common actions, including click, fill, and navigate
- Assertions
- Example test scenarios on a sample web application
- Shared lab introduced

## Day 3: Lab & Advanced Scenarios

**Target:** Continue the shared Playwright lab with in-class support, then review selected solutions and discuss advanced scenarios.

### Topics Covered

- Shared lab continued from Day 2: writing Playwright test scenarios
- In-class lab time with instructor support available
- More advanced Playwright scenarios and techniques
- Group review and discussion of selected lab work
- Wrap-up and next steps

## How to Turn In the Labs

Labs are done on your own branch of the classroom repo. When finished, push your branch and open a pull request into **main**.

Branch naming: **<lastname>-typescript-lab** or **<lastname>-playwright-lab**.

```
git checkout main && git pull
```

```
git checkout -b <lastname>-typescript-lab
```

```
git add .
```

```
git commit -m "Complete TypeScript lab"
```

```
git push -u origin <lastname>-typescript-lab
```

(Swap in **<lastname>-playwright-lab** and the matching commit message for the Playwright lab.)

Then open a PR on GitHub from your branch into **main**.

## Daily Class Flow

Each class session is four hours. Participants have time for review, guided instruction, breaks, and heads-down coding

|                        |                                                                                |
|------------------------|--------------------------------------------------------------------------------|
| <b>Opening Block</b>   | Review, questions, and introduction to the day's lecture                       |
| <b>Lecture Block 1</b> | First instructional section with examples and discussion                       |
| <b>Break</b>           |                                                                                |
| <b>Lecture Block 2</b> | Second instructional section with continued examples and guided practice       |
| <b>Lab Block</b>       | Final hour reserved for lab time and heads-down coding with instructor support |